GenAI

PROBLEM SOLVING

COURSE SLIDES · WSQ

WSQ - Generative AI for Problem Solving

WSQ Course Code: TGS-2023036653

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v21.0 · 17 August 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

ADMIN

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and for funding.

ICE BREAKER

Let's know each other.

Your name · your role · one workplace problem that keeps coming back no matter how many times you fix it.

About the Trainer

TN

Trainer Name

WSQ Accredited Adult Educator · Tertiary Infotech Academy

QUALIFICATIONS

INDUSTRY EXPERIENCE

SPECIALISATION

CONTACT

YOUR TRAINER TODAY

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer & Courseware Developer · Tertiary Infotech Academy

QUALIFICATIONS

PhD (NUS) · MEng (NTU) · MBA · ACTA · DACE

INDUSTRY EXPERIENCE

25+ years in engineering, data science, AI and process improvement

SPECIALISATION

Generative AI, problem solving, Six Sigma / SPC, design thinking

CONTACT

enquiry@tertiaryinfotech.com · +65 6100 0613

Ground Rules

1. **Phones on silent**
   Set your mobile phone to silent mode. Step out quietly for calls.

2. **Participate actively**
   No question is stupid. The best problem solvers ask the most obvious questions.

3. **Respect every view**
   Disagree with the idea, never the person. Diversity of view is how blind spots get found.

4. **Breaks**
   Exit and re-enter quietly during the session for toilet or phone breaks.

BEFORE WE START

Download Your Course Material

All slides, the Learner Guide and the activity briefs are on the LMS portal.

lms-tms.tertiaryinfotech.com

Generative AI for Problem Solving

- Trainer Slides (PPT)
- Learner Slides (PDF)
- Learner Guide (PDF)
- Activities Folder

1. Log in to https://lms-tms.tertiaryinfotech.com
2. Open My Courses and select this course
3. Click the Courseware tab
4. Download the slides, Learner Guide and activities

Skills Framework Alignment

1 **TSC Title**
Problem Identification

2 **TSC Code**
RET-ACE-4006-1.1

3 **Proficiency Level**
Level 4 — analyse causes, evaluate options and decide

4 **Course Code**
TGS-2023036653

SKILLS FRAMEWORK

TSC Abilities Covered

- A1: Identify the types of performance deficiency and examine the causes and their impact on organisation-related aspects
- A2: Identify the root causes of the problems with team members using appropriate group facilitation techniques
- A3: Deduce relevant linkages and patterns to identify key implications on organisational systems and processes
- A4: Develop corrective action plans for any shortfalls identified in implemented solutions
- A5: Shortlist and evaluate the most viable ideas using appropriate problem-solving and decision-making techniques and tools
- A6: Determine a preferred solution using appropriate methods and draw up implementation plans
- A7: Evaluate the effectiveness of the implemented solution and implementation plans

SKILLS FRAMEWORK

TSC Knowledge Covered

- K1: Criteria for identifying performance deficiency or cause of failure in organisational systems and processes
- K2: Types of analytical tools and techniques in terms of problem identification
- K3: Application of problem solving tools and techniques
- K4: Techniques used during problem solving and decision making processes
- K5: Group facilitation techniques for root cause identification
- K6: Types of decision making models for arriving at the preferred solution and their features
- K7: Techniques to evaluate the effectiveness of implemented solution and implementation plan
- K8: Factors affecting the effectiveness of an implementation plan
- K9: Rationale for the different components in a corrective action plan

BEFORE YOU START

Prerequisites

1

Level

Beginner — no prior problem-solving or AI training required.

2

Education

Minimum 3 GCE 'O' Level passes including English, or WPL Level 5.

3

Experience

At least 1 year of working experience.

4

Equipment

A laptop with a browser. Spare units available on request.

BY THE END OF THIS COURSE

Learning Outcomes

- LO1: Identify performance deficiencies and root causes impacting organisational aspects
- LO2: Deduce key implications, develop corrective plans, and shortlist viable solutions
- LO3: Determine preferred solutions and evaluate the effectiveness of implemented plans

Topic 1: Identifying Performance Gaps and Root Causes with Generative AI

1. A problem well stated is a problem half solved — poor problem definitions burn budget on the wrong fix.
2. Distinguish the symptom, the problem and the root cause; only root causes stop recurrence.
3. Well-defined vs ill-defined problems demand different tooling and different levels of framing effort.
4. A workplace problem statement carries Context, Metric, Baseline, Target, Timeframe and Constraints.
5. Root-cause toolkit: 5 Whys (depth), Fishbone (breadth), Pareto (priority), System Loops (dynamics).
6. GenAI accelerates framing and hypothesis generation, but the human owns evidence and judgement.

Topic 2: Developing Corrective Plans and Evaluating Potential Solutions

1. Separate divergence from convergence — generating and judging at the same time kills good ideas.
2. GenAI is a divergence engine: it produces breadth and unfamiliar analogies at near-zero cost.
3. Convergence needs explicit criteria — Impact vs Ease for speed, weighted decision matrix for rigour.
4. A corrective action plan states the root cause addressed, action, owner, timeline, resource and measure.
5. Cognitive fixedness and consensus fatigue are the two biases that quietly narrow the solution set.
6. Generative optimisation: GenAI explores the option space, structured criteria evaluate against constraints.

Topic 3: Selecting, Implementing and Measuring Solution Effectiveness

1. A preferred solution is a decision you can defend — criteria, evidence, trade-offs and risks made explicit.
2. An implementation plan converts a chosen solution into action, owner, timeline, resource and checkpoint.
3. Most solutions fail on adoption, not on design — stakeholder resistance is a design input, not a surprise.
4. Leading indicators warn you early; lagging indicators confirm the outcome. You need both.
5. Evaluate against the baseline in the problem statement — this closes the loop back to Topic 1.
6. PDCA / DMAIC-Control sustains the gain; without a control plan improvements decay.

SCHEDULE · 8 INSTRUCTIONAL HOURS

Lesson Plan — Day 1

Time	Session
9.30am	Digital Attendance (AM) · Introductions · Learning Outcomes
10.00am	Topic 1: Problem framing and measurable problem statements
11.00am	Activity 1 — Problem Statement (ShopFront SG)
12.30pm	Lunch Break (1 hour)
1.30pm	Digital Attendance (PM) · Root cause analysis toolkit
2.00pm	Activity 2 — 5 Whys (Meridian Health)
3.00pm	Activity 3 — Fishbone (Horizon Bank)
4.15pm	Activity 4 — Pareto (Nexa Logistics)
5.15pm	Activity 5 — System Loops (Horizon Bank revisited)
6.30pm	End of Day 1

WHEN IT MATTERS Frame the problem and find the true root cause — with GenAI as your analyst

SCHEDULE · 8 INSTRUCTIONAL HOURS

Lesson Plan — Day 2

Time	Session
9.30am	Digital Attendance (AM) · Recap of Day 1
10.00am	Topic 2: Divergent ideation and GenAI as a divergence engine
10.45am	Activity 6 — Divergent Ideation + SCAMPER (Meridian Health)
11.45am	Activity 7 — Impact-Ease and weighted decision matrix
12.30pm	Lunch Break (1 hour)
1.30pm	Digital Attendance (PM) · Activity 8 — Corrective Action Plan
2.30pm	Topic 3: Implementation, change management and evaluation
3.00pm	Activity 9 — Implementation & stakeholder resistance (ShopFront SG)
4.00pm	Activity 10 — Measuring effectiveness at 90 days
4.45pm	Revision · Course feedback · TRAQOM survey
5.15pm	Digital Attendance (Assessment) · Final Assessment (WA + Case Study)
6.30pm	End of Course

WHEN IT MATTERS Generate, choose, implement and prove the solution — with GenAI as your consultant

READ BEFORE THE ASSESSMENT

Briefing for Assessment

1 **Clear your table**
Place phones and all other materials under the table or on the floor.

2 **No photos or recording**
Assessment scripts must not be photographed, copied or recorded.

3 **No discussion**
The assessment is individual work. No discussion once the paper is handed out.

4 **Use black or blue pen**
For hard-copy scripts. No liquid paper or correction tape.

5 **Time is called**
Scripts are collected when time is up. Late submissions are not accepted.

6 **Open book means**
Slides, Learner Guide and approved materials only — no internet, no AI tools.

Final Assessment

1

Written Assessment (WA)

Short-answer questions covering problem identification, root cause analysis, solution selection and evaluation.

2

Case Study (CS)

An integrated workplace scenario — you diagnose, decide and justify, exactly as in the activities.

3

Open book

Slides, Learner Guide and approved materials only. No internet search, no AI tools during assessment.

4

Competent / Not Yet Competent

Both instruments must be assessed Competent. A re-assessment path is available.

Assessment Flow

1. **TRAQOM Survey** — Complete on the LMS
2. **Digital Attendance** — Assessment QR scan
3. **Written Assessment** — SAQ · 1 hour
4. **Case Study** — Scenario · 1 hour
5. **Submit on LMS** — Upload your answers
6. **Summary Record** — Sign the ASR

RE-ASSESSMENT

A learner assessed Not Yet Competent is counselled on the gap and offered a re-assessment. An appeal may be lodged with the training administrator within 7 working days.

MANDATORY ADMIN

Digital Attendance & TRAQOM Survey

1

AM · PM · Assessment

Digital attendance is taken three times a day. It is mandatory for every WSQ-funded course.

2

Scan the SSG QR code

The trainer displays the QR code from the SSG portal. Scan it with your phone camera and submit.

3

75% minimum attendance

Below 75% you are not eligible for assessment or for course fee funding.

4

TRAQOM survey

Complete the mandatory post-course survey and collect your certificate at https://lms-tms.tertiaryinfotech.com

Criteria for Course Fee Funding

1

Attendance

Minimum 75% attendance based on the SSG Digital Attendance record.

2

Assessment

Complete both instruments and be assessed 'Competent'.

3

Certificate

Certificate of Achievement from Tertiary Infotech Academy Pte Ltd.

4

SSG Statement

OpenCert / Statement of Achievement issued by SkillsFuture Singapore.

01

TOPIC 1

Identifying Performance Gaps and Root Causes with Generative AI

Problem framing · Problem statements · 5 Whys · Fishbone · Pareto · System loops

WHY FRAMING COMES FIRST

A problem well stated is a problem half solved.

Charles Kettering. Most organisations skip straight to solutions — and pay for it twice.

What Problem Solving Actually Is

1. **Issue identification**
   Is the problem familiar or novel, simple or complex? Different answers demand different tools.

2. **Information gathering**
   Collect the facts. Find out how similar problems were solved before.

3. **Analysis**
   Break the problem into parts and establish cause and effect.

4. **Action generation**
   Develop several possible solutions across short and long horizons.

5. **Implementation**
   Select and execute the most promising approach — and stay flexible.

6. **Evaluation**
   Assess whether it worked, and extract the lesson for next time.

The IDEAL Problem-Solving Cycle

1. **Identify** — the real problem
2. **Define** — the context
3. **Explore** — possible strategies
4. **Act** — on the best option
5. **Look back** — and learn

THE DISCIPLINE

Most teams start at 'Act'. The cost of skipping Identify and Define is paid later, with interest — you fix the wrong thing, then fix it again.

THE DISTINCTION THAT SAVES BUDGETS

Symptom vs Problem vs Root Cause

	Symptom	Problem	Root Cause
What it is	What people notice	The measurable deficiency	The system condition producing it
Example	'Customers are unhappy'	Checkout completion fell 68%→51%	No capacity planning from analytics
Fixing it	Feels responsive	Improves the number	Stops recurrence
If you stop here	Problem returns next quarter	Returns in another form	It is actually solved

WHEN IT MATTERS Treating a symptom always feels faster. It is the most expensive habit in problem solving.

DIFFERENT PROBLEMS, DIFFERENT TOOLS

Well-Defined vs Ill-Defined Problems

	Well-defined	Ill-defined
Goal	Clear and specific	Vague or contested
Path	Known method exists	No standard method
Example	Reduce load time to under 2s	Improve the customer experience
Right approach	Apply the known technique	Frame it first, then decompose
GenAI's role	Execute and check	Explore, reframe, generate options

WHEN IT MATTERS Most workplace problems arrive ill-defined. Framing IS the work.

THE TEMPLATE YOU WILL USE ALL COURSE

Six Elements of a Workplace Problem Statement

1. **Context**
   Who, where, and which process is affected.

2. **Metric**
   The single primary measure you will move.

3. **Baseline**
   Where that metric is today — with the number.

4. **Target**
   Where it must be — with the number.

5. **Timeframe**
   By when. Without this, nothing is ever late.

6. **Constraints**
   Budget, headcount, compliance, brand — what you may not change.

COMPARE

Good or Bad Problem Definition?

✗ BAD

- "Our customers are not happy with our services."
- No metric. No baseline. No target. No timeframe.
- Every department will interpret it differently.
- Unmeasurable — you can never prove it was solved.

✓ GOOD

- "Checkout completion on our app fell from 68% to 51% over 12 months against a 70% benchmark. We aim to restore it to 68% within 90 days without added headcount."
- Metric, baseline, target, timeframe and constraint all present.
- Actionable by a specific team.
- Evaluable at day 90 against a real number.

Barriers That Quietly Wreck Problem Solving

1 **Functional fixedness**
Seeing a thing only for its usual purpose — the reason obvious re-uses stay invisible.

2 **Mental set**
Re-using the solution that worked last time, on a problem that has changed.

3 **Confirmation bias**
Collecting evidence that supports the cause you already suspect.

4 **Unnecessary constraints**
Self-imposed limits nobody actually stated. Most 'fixed' constraints are habits.

5 **Irrelevant information**
Rich data that distracts from the few numbers that matter.

6 **Consensus fatigue**
Accepting a weak option just to end the meeting.

SETTING EXPECTATIONS

GenAI does not solve your problem.

It expands what you consider, and accelerates how fast you get there. The judgement stays yours.

What GenAI Is Good At — and Not

Dimension	GenAI is strong	GenAI is weak
Breadth of options	Generates 20 ideas in seconds, free of politics	Cannot tell which are real
Framing help	Restates vague problems into structured form	Invents specifics not in your evidence
Analogy	Borrows patterns across every industry	May transfer an inapplicable pattern
Arithmetic	Improving, but still errs	Verify every number it produces
Context	Applies general best practice	Does not know your people, history or politics
Accountability	Tireless, unbiased by hierarchy	Cannot be held responsible for the decision

WHEN IT MATTERS Use it as a divergence engine and a challenger — never as an oracle.

WHERE IT ADDS VALUE

How GenAI Fits the Problem-Solving Cycle

1. **Frame** — Sharpen the statement
2. **Diagnose** — Generate cause hypotheses
3. **Ideate** — Expand the option set
4. **Decide** — Score against your…
5. **Evaluate** — Structure the evidence

THE HUMAN'S JOB

Supply the evidence, challenge the assumptions, own the decision. The AI proposes; you dispose.

Prompting Habits That Actually Work

1. **Give it a role**
   'Act as a root cause analysis expert' sets the reasoning frame.

2. **Supply the evidence**
   Paste your real numbers. Without them the AI invents plausible ones.

3. **Demand structure**
   Name the fields you want back — Context, Metric, Baseline, Target.

4. **Ask it to flag assumptions**
   'Mark anything you assumed that my evidence did not support.'

5. **Make it challenge you**
   'Ask me for evidence rather than accepting my answer.'

6. **Iterate, don't accept**
   The first output is a draft. The third is usually useful.

TOPIC 1 · PROMPT 1

The Problem Statement Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
You are a business analyst for an e-commerce company.

Refine this vague problem into a clear, measurable problem statement:
<<State your problem here>>

Include, each clearly labelled:
- Context (who, where, which process)
- Metric (the single primary measure)
- Baseline (where it is now, with the number)
- Target (where it must be, with the number)
- Timeframe (by when)
- Constraints (budget, headcount, compliance)

Then list every assumption you had to make, and state what
evidence would let me remove that assumption.

Do not propose solutions yet.
```

WATCH FOR The AI will supply specifics your evidence never contained. Mark each one as an assumption.

Activity 1 — Rewriting a Vague Problem into a Measurable Problem Statement

ACTIVITY 1

A1 · K1 — Identify the type of performance deficiency and express it as a measurable problem statement.

ShopFront SG — 'Customers are unhappy and sales are dropping'

Rewrite an executive's vague complaint into a measurable problem statement your team can defend.

TEAM

Teams of 3

TIME

45 minutes

TOOL

CollabNote · https://alfredang.github.io/collabnote/

The Situation

CASE STUDY

ShopFront SG — 'Customers are unhappy and sales are dropping'

ShopFront SG is a 42-outlet fashion retailer with an e-commerce site and a mobile app. At the Monday management meeting the Head of Retail Operations opens with one line: "Customers are unhappy and sales are dropping — fix it." No metric, no baseline, no scope. Three departments leave the room with three different interpretations. Marketing books a S$180,000 discount campaign. IT starts an app redesign. Customer Service hires two temps. Ninety days later the complaint rate is unchanged and the budget is gone.

You are the business analyst asked to restate the problem BEFORE any more money is committed. The dashboard extract below is all the evidence you have.

The Evidence

Metric	12 months ago	Current	Industry benchmark
Average app page load	2.1 s	4.8 s	under 2.0 s
Checkout completion rate	68%	51%	70%
Cart abandonment	61%	74%	60%
CSAT (post-purchase)	84%	70%	85%
Repeat purchase (90-day)	38%	29%	40%
Complaint tickets / week	120	410	-
In-store sales	S$4.2m/mo	S$4.1m/mo	-
Online sales	S$2.8m/mo	S$1.9m/mo	-

WHEN IT MATTERS ShopFront SG — 90-day performance extract

ACTIVITY 1 · RUN THIS

Activity 1 — The Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
You are a business analyst for a Singapore fashion retailer.

Refine the vague statement below into ONE clear, measurable problem statement.

Vague statement: "Customers are unhappy and sales are dropping."

Evidence:
- App page load rose from 2.1s to 4.8s (benchmark <2.0s)
- Checkout completion fell 68% -> 51% (benchmark 70%)
- Cart abandonment rose 61% -> 74%
- CSAT fell 84% -> 70%
- Online sales fell S$2.8m -> S$1.9m per month; in-store flat

Your problem statement MUST contain, each labelled:
- Context (who, where, which process)
- Metric (the single primary measure)
- Baseline (where it is now, with the number)
- Target (where it must be, with the number)
- Timeframe (by when)
- Constraints (budget, headcount, compliance, brand)

Then list any assumption you had to make, and state what additional evidence would let you remove that assumption. Do not propose solutions yet.
```

ACTIVITY 1 · BREAKOUT

Activity 1 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Which metric in the table is the SYMPTOM the executive noticed, and which metric is closest to the actual performance deficiency? Justify with the numbers.

Q2 The evidence shows in-store sales are flat while online sales fell 32%. What does that single contrast rule out as a cause, and why does that matter before you spend a dollar?

Q3 Write the problem statement with all six elements. Which element was hardest to fill, and what does the gap tell you about your organisation's data?

Q4 Compare your team's statement with the GenAI-generated one. What did the AI assume that is NOT supported by the evidence? (Every team will find at least one.)

Q5 Whose sign-off do you need on this problem statement before work starts, and what would you show them to get it?

Activity 1 — Debrief

TRAINER DEBRIEF

THE DEFICIENCY

In-store flat while online fell 32% localises the problem to the digital funnel — which invalidates the S$180k discount campaign before a dollar more is spent.

SYMPTOM VS PROBLEM

CSAT is a lagging perception measure. Checkout completion is the operational deficiency you can act on.

THE AI'S ASSUMPTIONS

Every team finds the AI inventing peak-hour windows, gateway names or segments the evidence never supplied. Mark them. This is the core AI-literacy habit.

WHAT IS STILL UNKNOWN

WHY load time rose is root cause — that is Activity 2. Resist 'we need more servers'.

THE ROOT CAUSE TOOLKIT

Four tools. Four different jobs.

Depth · Breadth · Priority · Dynamics. Mature problem solvers use them together, not instead of each other.

Choosing Your Root Cause Tool

Tool	What it gives you	Use it when	Limitation
5 Whys	Depth on one causal chain	The problem is focused	Only follows one branch
Fishbone	Breadth across all dimensions	Causes are many and unclear	Shallow on each
Pareto	Priority by contribution	You have counted data	Ranks frequency, not severity
System Loops	Feedback dynamics over time	Fixes keep rebounding	Needs time-series insight

WHEN IT MATTERS Pareto tells you WHERE to look. 5 Whys tells you WHY. They chain together.

Ed-Tool: 5 Whys

5 Whys Analysis Tool

https://alfredang.github.io/5whys/

WHAT IT DOES

Drives a symptom down to a systemic root cause by asking 'why' five times, recording each level so the chain stays visible and auditable.

HOW YOU'LL USE IT

Enter the problem statement, then add one why and answer per level. Stay on ONE causal branch — depth is the point.

5 Whys — Worked Example

1. **Why do morning appointments overrun?**
   Blood draws take longer than the slot allows.

2. **Why do draws take longer?**
   40% of morning staff are relief at 14 min vs 6 min; 18% need a repeat.

3. **Why so many relief staff?**
   Only 3 of 5 permanent phlebotomists are rostered at 07:00.

4. **Why under-rostered at 07:00?**
   Turnover is 44%/yr; vacancies are backfilled by untrained agency staff.

5. **Why is turnover 44%?**
   The early shift is unattractive and there is no structured onboarding or career path.

ROOT CAUSE IDENTIFIED

A workforce retention and onboarding failure — surfacing as a capacity problem.

WHY ROOT CAUSE PAYS

"We're short-staffed" was wrong in the way that matters.

They are not short of headcount. They are short of TRAINED headcount at 07:00. That distinction is worth S$1.2m.

TOPIC 1 · PROMPT 2

The 5 Whys Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a root cause analysis expert facilitating a clinical operations team.

Problem: 31% of morning health-screening appointments at Meridian Health overrun by more than 40 minutes, and 23% of blood samples miss the fixed 09:45 lab courier.

Evidence: only 3 of 5 phlebotomists are rostered at 07:00 (all 5 by 08:30); relief staff take 14 min per draw vs 6 min for trained staff; relief staff are 40% of the morning roster; 18% of draws need a repeat attempt; permanent staff turnover is 44% a year.

Apply the 5 Whys technique. Ask ONE why at a time and wait for my answer before asking the next. Go five levels deep.

At each level, state what EVIDENCE would confirm or refute that level — do not accept my answer at face value. If my answer is an opinion rather than a fact, say so and ask me for the data that would support it.

After level 5, summarise: the root cause, the evidence supporting it, and which levels remain unverified assumptions.
```

WATCH FOR Instructing the AI to demand evidence at each level turns it into a facilitator, not an echo.

Activity 2 — 5 Whys Root Cause Analysis

ACTIVITY 2

A2 · K2, K5 — Identify root causes with team members using structured group facilitation.

Meridian Health Screening — the 07:00 blood-test backlog

Drive a staffing complaint down five levels to the systemic cause — and test a S$1.2m proposal against it.

TEAM

Teams of 3-5

TIME

45 minutes

TOOL

5 Whys Tool · https://alfredang.github.io/5whys/

ACTIVITY 2 · CASE STUDY

The Situation

CASE STUDY **Meridian Health Screening — the 07:00 blood-test backlog**

Meridian Health operates six health-screening centres in Singapore. Corporate clients book annual screening packages; the fasting blood draw must happen between 07:00 and 09:30.

Over the last four months, 31% of morning appointments overrun by more than 40 minutes. Corporate clients have escalated twice. One major client (1,400 employees, S$310,000 annual contract) has given notice of review. The Operations Manager's proposal is to open a seventh centre at a capital cost of S$1.2m.

The Clinical Director is unconvinced. She has asked your team to establish the ROOT CAUSE before the board approves any capital expenditure. The nurses tell you the same thing every morning: "We're just short-staffed." The data below says something more interesting.

ACTIVITY 2 · DATA

The Evidence

Observation	Value
Booked slots 07:00-09:30	48 per centre
Phlebotomists rostered 07:00	3 of 5
Phlebotomists rostered 08:30	5 of 5
Mean draw time, trained staff	6 min
Mean draw time, relief staff	14 min
Relief staff share of morning roster	40%
Repeat draws (failed first attempt)	18%
Clients arriving without fasting compliance	12%
Lab courier pickup (fixed)	09:45
Samples missing the 09:45 courier	23%
Permanent staff turnover, 12 months	44%

WHEN IT MATTERS Meridian Health — morning screening operations, 4-month sample

Activity 2 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Run the 5 Whys to five levels. At which level did you stop describing WHAT happens and start explaining WHY the system produces it?

Q2 The nurses say 'we're short-staffed'. The data shows all 5 phlebotomists are rostered by 08:30. What is the real constraint — headcount, or something else?

Q3 Trace the causal chain to 44% annual turnover. If turnover is the root cause, what does that say about the S$1.2m seventh centre proposal?

Q4 Where could this 5 Whys chain have gone wrong? Identify one point where a different, equally plausible 'why' would have led you somewhere completely different.

Q5 The GenAI facilitator challenged at least one of your answers as opinion rather than fact. Which one, and were you able to supply the evidence?

Activity 2 — Debrief

TRAINER DEBRIEF

THE CHAIN

Overrun → relief staff at 14 min → only 3 of 5 permanent rostered → 44% turnover → unattractive early shift with no onboarding or career path.

THE S$1.2M TEST

A seventh centre staffed by the same untrained relief pool reproduces the same overrun. You buy the problem twice.

FACILITATION DISCIPLINE

One why at a time; every level must survive 'what evidence supports this?'. The AI modelling that challenge is the K5 outcome.

THE BRANCH POINT

At Why 2 a team could have followed the 12% fasting non-compliance instead. That is a legitimate second cause — and exactly why we use Fishbone next.

BREADTH ACROSS ALL DIMENSIONS

The Fishbone (Ishikawa) Diagram

1

Who invented it

Dr Kaoru Ishikawa, Japanese quality control expert — to stop teams treating symptoms.

2

What it does

Categorises every potential cause of one problem so nothing is missed.

3

The head

The problem statement — measurable, with the number.

4

The bones

People · Process · Technology · Material/Information · Environment · Measurement.

5

The twigs

Specific candidate causes on each bone, phrased as causes, not complaints.

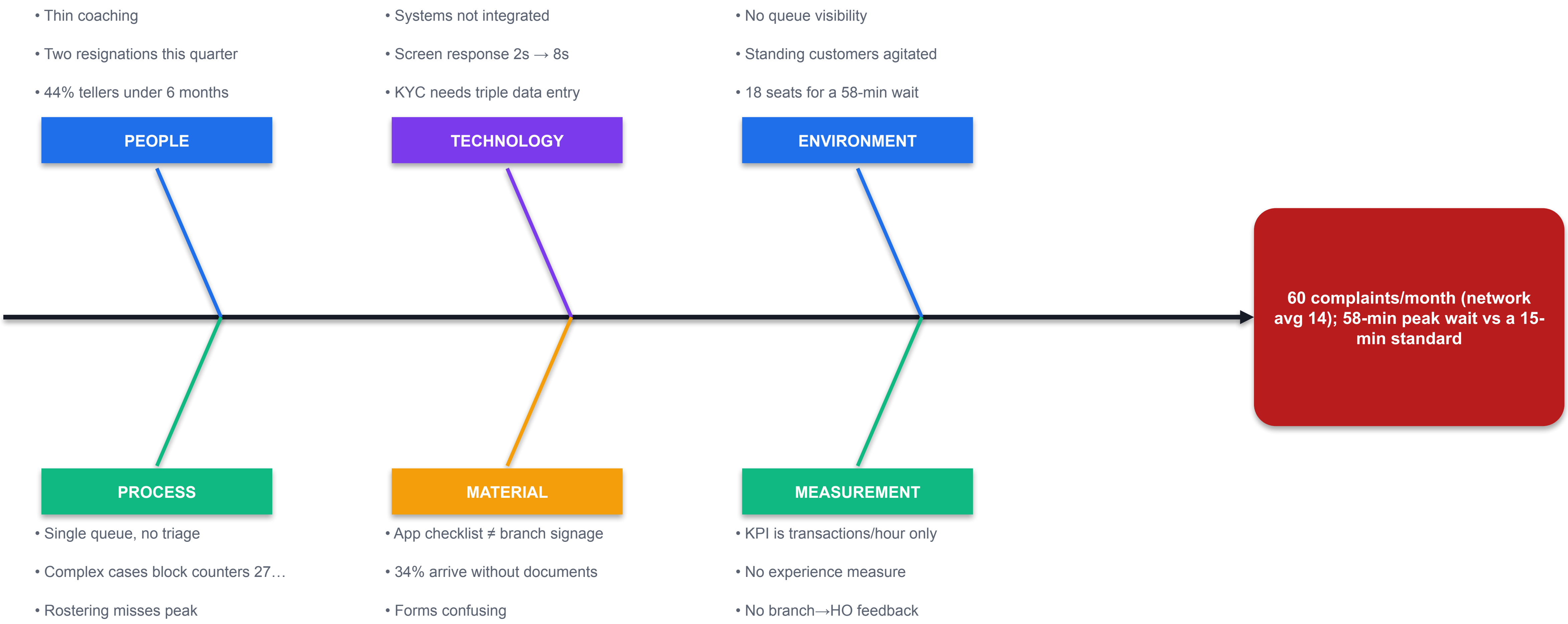
6

The payoff

Shows you which dimension your intervention is actually targeting.

SIX BONES · EVIDENCE-LOADED

Fishbone — Horizon Bank Jurong East



THE EXPENSIVE LESSON

Head Office trained the People bone.

It carries the fewest evidenced causes. That is precisely why complaints did not move — they intervened on the wrong bone.

Ed-Tool: Fishbone

Fishbone Diagram Tool

https://alfredang.github.io/fishbone/

WHAT IT DOES

Builds an Ishikawa diagram across six cause categories, so a team can see at a glance which dimension carries the weight of evidence.

HOW YOU'LL USE IT

Enter the problem as the head, then add causes bone by bone. Mark each [EVIDENCED] or [HYPOTHESIS].

TOPIC 1 · PROMPT 3

The Fishbone Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a quality improvement facilitator running an Ishikawa (fishbone) analysis with a bank branch team.

Problem (the fish head): Customer complaints at the Horizon Bank Jurong East branch have risen to 60 per month (network average 14), with peak
counter wait time at 58 minutes against a 15-minute standard.

Generate candidate causes under these six bones: People, Process, Technology, Material/Information, Environment, Measurement.

Rules:
- Give 4-6 candidate causes per bone, each phrased as a CAUSE not a complaint
- Mark each cause [EVIDENCED] or [HYPOTHESIS] based only on what I give you below
- Do NOT propose solutions
- Finish by naming the 3 causes you would investigate first and say why

Evidence available: single queue with no triage; 61% of footfall between 11:30-14:00; 3.2 of 6 counters staffed at peak; new KYC platform
requires triple data entry; screen response degraded 2s to 8s; 44% of tellers under 6 months tenure; teller KPI is transactions-per-hour only;
34% of customers arrive without documents; app checklist and branch signage disagree; complex cases take 27 minutes at the counter; no branch-
to-HO feedback route; 18 seats in the waiting area.
```

WATCH FOR Forcing [EVIDENCED] vs [HYPOTHESIS] labels stops the AI's confident guesses becoming your facts.

Activity 3 — Fishbone (Ishikawa) Diagram for Breadth of Causes

ACTIVITY 3

A2 · K2, K5 — Categorise potential causes across all contributing dimensions.

Horizon Bank — branch service complaints at Jurong East

Map every contributing cause across six dimensions — and discover why the training refresher failed.

TEAM

Teams of 3-5

TIME

45 minutes

TOOL

Fishbone Tool · https://alfredang.github.io/fishbone/

ACTIVITY 3 · CASE STUDY

The Situation

CASE STUDY

Horizon Bank — branch service complaints at Jurong East

Horizon Bank's Jurong East branch has moved from best to worst in the retail network on customer experience in eleven months. Complaints run at 60 per month against a network average of 14. Average counter wait time is 58 minutes at peak; the service standard is 15.

Head Office ran a 'Service Excellence' refresher for all branch staff four months ago. Complaints did not move. The Branch Manager is now under performance review and morale is visibly poor — two tellers have resigned this quarter.

The Regional Director suspects the training was aimed at the wrong cause. Your team has been asked to map ALL contributing causes across every dimension before another intervention is funded. 5 Whys gave depth on one branch; this problem is too broad for one branch.

ACTIVITY 3 · DATA

The Evidence

Dimension	Observation
Queue	Single queue for all transaction types; no triage between simple and complex
Peak	11:30-14:00 accounts for 61% of daily footfall (lunch crowd from nearby offices)
Counters	6 counters exist; mean 3.2 staffed at peak
Systems	New KYC platform launched 11 months ago; teller must key client data into 3 systems
Systems	Mean core-banking screen response 8s, was 2s before the KYC platform
Staff	44% of tellers have under 6 months tenure
Staff	Teller KPI is transactions-per-hour; no customer-experience measure
Customer	34% of customers arrive without required documents
Customer	Branch signage and the app's document checklist do not match
Process	Complex cases (loans, disputes) occupy a counter for a mean of 27 minutes
Management	No structured feedback route from branch to Head Office product owners
Environment	Only 18 seats in the waiting area; standing customers visibly agitated

WHEN IT MATTERS Horizon Bank Jurong East — observation log and system data

Activity 3 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Populate all six bones. Which bone filled up fastest, and which was hardest? What does an empty bone usually mean — no causes, or no visibility?

Q2 Head Office's answer was a Service Excellence training refresher. Which bone does that intervention target, and what share of your evidenced causes sit on that bone?

Q3 Find a cause on your diagram that CREATES another cause on a different bone. (Hint: look at the KPI and the queue.) What does that tell you about treating bones as independent?

Q4 Which three causes would you investigate first? Defend the ranking on evidence strength and likely contribution — not on ease of fixing.

Q5 The teller KPI is transactions-per-hour. Predict the behaviour that KPI produces at a counter when a confused elderly customer needs 20 minutes. Which bone does that belong on?

Activity 3 — Debrief

TRAINER DEBRIEF

WHERE THE EVIDENCE SITS

Process, Technology and Measurement carry the weight. People carries the least — and People is exactly what Head Office trained.

CROSS-BONE CAUSATION

The transactions/hour KPI drives tellers to rush complex cases back into the queue, lengthening waits. Bones are not independent.

FISHBONE VS 5 WHYS

One deep chain vs six shallow ones. Depth without breadth fixes one cause and the problem persists; breadth without depth fixes symptoms everywhere.

GENAI'S ROLE

Excellent at populating bones fast; poor at knowing which are real. Count how many came back [HYPOTHESIS] — those are your verification list.

PRIORITY · WHERE TO LOOK FIRST

Pareto Analysis — the 80/20 Rule

1 **The principle**
Roughly 80% of the effect comes from roughly 20% of the causes.

2 **What it does**
Breaks a big problem into ranked pieces so effort lands where it counts.

3 **The vital few**
The small set of categories carrying most of the volume.

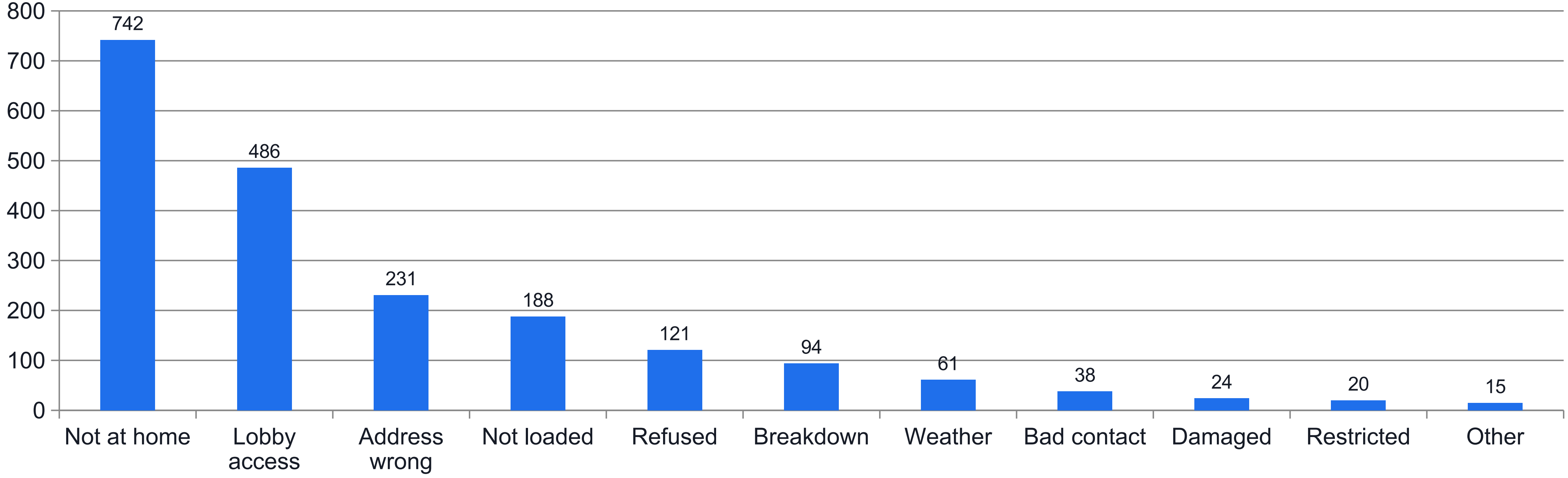
4 **The trivial many**
The long tail — real, but not where your first dollar goes.

5 **Ranking dimension**
Count, cost, risk or customer harm. Choosing this IS the analysis.

6 **The limitation**
It ranks frequency. Frequency is not severity.

80/20 IN PRACTICE

Pareto — Nexa Logistics Delivery Failures



WHAT THE DATA SHOWS

Four of eleven reasons carry 82% of failures; the top two alone carry 61%. 'Address incorrect' — the one generating the angriest emails — is only 11.6%. Loud is not the same as large.

SHOW YOUR WORKING

A 60% cut on the top two still misses the SLA.

1,228 × 0.6 = 737 removed, leaving 1,263 of 40,000 = 3.2%. The 2% SLA needs the top four. Always do the arithmetic.

TOPIC 1 · TOOL

Ed-Tool: Pareto Chart

Pareto Chart Tool

https://alfredang.github.io/paretochart/

WHAT IT DOES

Sorts categories by count, computes the cumulative percentage line, and shows exactly where the 80% cut-off falls.

HOW YOU'LL USE IT

Enter each failure reason and its count. Read the cumulative line to find your vital few.

TOPIC 1 · PROMPT 4

The Pareto Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as an operations analyst. Perform a Pareto (80/20) analysis on the delivery failure data below for a Singapore last-mile
logistics company.

Total deliveries: 40,000/month. Failures: 2,000 (5%). SLA target: 2%. Re-attempt cost: S$8.50 each.

Data (reason, count):
Recipient not at home 742; Lobby access denied 486; Address incorrect 231; Parcel not loaded 188; Recipient refused 121; Vehicle
breakdown 94; Weather 61; Invalid contact 38; Damaged 24; Restricted hours 20; Other 15.

Produce:
1. A table with count, percentage, and CUMULATIVE percentage, sorted descending
2. The cut-off line — which reasons make up the vital few (roughly 80%)
3. The monthly S$ recovery if the top 2 reasons were reduced by 60%
4. Whether hitting the 2% SLA is achievable by fixing the vital few alone — show the arithmetic
5. One caution about what this Pareto analysis does NOT tell me
```

WATCH FOR Always verify the AI's percentages against the tool. Arithmetic is where it still slips.

Activity 4 — Pareto Analysis to Target the Vital Few

ACTIVITY 4

A1, A3 · K2 — Prioritise causes by contribution and identify key implications.

Nexa Logistics — 2,000 failed deliveries a month

Rank eleven failure reasons, find the vital few, and decide where two funded initiatives go.

TEAM

Teams of 3-5

TIME

40 minutes

TOOL

Pareto Chart Tool · https://alfredang.github.io/paretochart/

ACTIVITY 4 · CASE STUDY

The Situation

CASE STUDY

Nexa Logistics — 2,000 failed deliveries a month

Nexa Logistics runs last-mile delivery for e-commerce clients across Singapore. Of 40,000 monthly deliveries, 2,000 fail on first attempt — a 5% failure rate against a contractual SLA of 2%. Every failed delivery costs S$8.50 to re-attempt: S$17,000 a month, and two clients have invoked SLA penalty clauses.

The Operations Director has a list of eleven failure reasons and a budget that will fund roughly two initiatives this quarter. His instinct is to start with 'Address incorrect' because it generates the angriest client emails.

Your team must decide where the two initiatives should go — using the data, not the noise.

ACTIVITY 4 · DATA

The Evidence

Failure reason	Count	Cost impact (S$)
Recipient not at home (no delivery window given)	742	6307
Driver could not access condo / office lobby	486	4131
Address incomplete or incorrect	231	1964
Parcel not loaded (sorting error)	188	1598
Recipient refused (wrong item expected)	121	1029
Vehicle breakdown / route disruption	94	799
Weather-related suspension	61	519
Recipient contact number invalid	38	323
Parcel damaged in transit	24	204
Restricted delivery hours at destination	20	170
Other / unclassified	15	128

WHEN IT MATTERS Nexa Logistics — first-attempt delivery failures by reason, last month

Activity 4 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Build the Pareto chart. How many of the eleven reasons account for roughly 80% of failures? What does that ratio mean for how the Operations Director should spend his budget?

Q2 The Director wants to start with 'Address incorrect' because it generates the angriest emails. What does the data say, and how would you make that argument to him without dismissing his concern?

Q3 Calculate: if the top two reasons drop by 60%, what is the new failure rate? Does that meet the 2% SLA? Show your arithmetic.

Q4 The top two reasons — 'not at home' and 'lobby access' — look like customer problems. Reframe each as something Nexa controls. What changes about who owns the fix?

Q5 Pareto ranks by FREQUENCY. Name a situation in your own workplace where the most frequent problem is not the most important one. What would you rank by instead?

Activity 4 — Debrief

TRAINER DEBRIEF

THE VITAL FEW

Four of eleven reasons = 82%. Top two = 61%. 'Address incorrect' is third at 11.6%.

LOUD VS LARGE

The Director responds to escalation volume, which tracks client temperament, not failure frequency. Pareto is how you have that conversation with evidence.

REFRAME OWNERSHIP

'Not at home' is a customer problem only if you never offered a delivery window. Restating moves 61% of the problem into Nexa's control.

WHAT PARETO CANNOT DO

It ranks frequency, never severity — and it tells you WHERE, never WHY. The 742 failures still need a 5 Whys.

Systems Thinking — Why Fixes Rebound

1

A system, not a queue

Organisations are webs of feedback loops, not linear cause-and-effect chains.

2

Reinforcing loop (R)

Compounds over time — makes things better and better, or worse and worse.

3

Balancing loop (B)

Pushes toward a target and stabilises — this is what most 'fixes' pull.

4

Delay

R-loops act slowly. That delay is why a bad decision can look right in month one.

5

Dormant loops

A balancing loop that exists but is not running is as dangerous as an active R-loop.

6

Leverage

Breaking one link in an R-loop beats adding effort to a B-loop forever.

TWO REINFORCING · TWO BALANCING

Horizon Bank — The Four Loops

B

B1 · BALANCING

Capacity control — what he intended

Wait time ↑
↳ Management pressure ↑
↳ Counters staffed ↑
↳ Service capacity ↑
↳ Wait time ↓

Real — and it produced the month-1 win

R

R1 · REINFORCING

Redeployment backfire

Counters staffed ↑
↳ Back-office staff ↓
↳ Backlog ↑
↳ Rework returns to counter ↑
↳ Handling time ↑ (9→13 min)
↳ Wait time ↑

Compounds — overwhelms B1 by month 4

R

R2 · REINFORCING

Burnout spiral

Workload ↑
↳ Overtime ↑ (88→196h)
↳ Resignations ↑
↳ Inexperience ↑ (44%→58%)
↳ Errors ↑ (6%→14%)
↳ Rework ↑

Compounds — feeds R1

B

B2 · BALANCING

Digital migration — DORMANT

Wait time ↑
↳ Willingness to try digital ↑
↳ Digital adoption ↑
↳ Branch volume ↓
↳ Wait time ↓

Never engaged — adoption flat at ~20%

THE DELAY TRAP

A quick win measured too early is indistinguishable from a mistake.

B1 acts in days. R1 and R2 act over months. The month-1 result validated a decision that was making things worse.

Leverage vs Effort

	Effort	Leverage
Example	Add more counters	Triage complex cases away from the counter
Cost	Must be sustained forever	Changes the structure once
Effect on R1	Feeds it	Breaks a link in it
Sustainability	Decays the moment you stop	Holds without continuous push
Test	'Are we pushing harder?'	'Have we changed what produces the behaviour?'

WHEN IT MATTERS If your answer is 'hire more people', you have described effort, not leverage.

TOPIC 1 · TOOL

Ed-Tool: System Loop

System Loop Tool

https://alfredang.github.io/systemloop/

WHAT IT DOES

Maps variables and causal links into closed feedback loops, marking each reinforcing (R) or balancing (B) so loop dynamics become visible.

HOW YOU'LL USE IT

Add variables as nodes, link them with (+)/(−), and close the loop. Count negative links: even = reinforcing, odd = balancing.

TOPIC 1 · PROMPT 5

The Systems Thinking Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a systems thinking facilitator. Help me map the causal feedback loops in this case.

Situation: A bank branch added 2 counters at peak by redeploying back-office staff. Wait time fell from 58 to 34 minutes in month
1, then rose to 61 minutes by month 4 — worse than the starting point.

Observed changes over 4 months: back-office backlog 0.5 -> 2.0 days; teller errors 6% -> 14%; rework returning to the counter
rose; overtime 88 -> 196 hrs/month; resignations rose; share of tellers under 6 months 44% -> 58%; mean handling time 9 -> 13 min;
digital adoption flat at ~20%.

Produce:
1. At least 2 REINFORCING (R) loops and 2 BALANCING (B) loops. For each, list the variables in causal order with (+) or (-) on
each link, and name the loop
2. Explain specifically WHY the month-1 improvement reversed — which loop overwhelmed which
3. Identify the delay in the system and explain why the delay made the wrong decision look right
4. Name 2 leverage points that would break the reinforcing loops, and say what makes them leverage rather than just more effort
```

WATCH FOR Ask explicitly for the DELAY and the dormant loop — the AI rarely volunteers either.

Activity 5 — System Loops: Why the Problem Keeps Coming Back

ACTIVITY 5

A3 · K2 — Deduce linkages and patterns to identify key implications on organisational systems.

Horizon Bank revisited — the fix that made it worse

Map the loops that turned a successful fix into a worse outcome, and find the real leverage points.

TEAM

Teams of 3-5

TIME

50 minutes

TOOL

System Loop Tool · https://alfredang.github.io/systemloop/

ACTIVITY 5 · CASE STUDY

The Situation

CASE STUDY

Horizon Bank revisited — the fix that made it worse

Return to Horizon Bank Jurong East. Acting on the fishbone, the Regional Director did the obvious thing: he ordered two additional counters staffed at peak, redeploying staff from the back office.

Month 1: average wait fell from 58 to 34 minutes. Everyone celebrated. Month 4: average wait is back to 61 minutes — WORSE than before. Complaints are at 68 per month. Three more tellers have resigned. Back-office processing is now two days behind, which has started generating a new complaint category of its own.

The Regional Director is baffled: "We added capacity and the problem came back bigger. What am I missing?"

What he is missing is that a branch is not a queue — it is a system of feedback loops. Your team must map them.

ACTIVITY 5 · DATA

The Evidence

Variable	Month 1	Month 4	Direction
Counters staffed at peak	3.2	5.1	up
Average wait time (min)	58 -> 34	61	down then up
Back-office processing backlog (days)	0.5	2.0	up
Teller errors requiring rework	6%	14%	up
Rework returning to the counter	low	high	up
Teller overtime hours / month	88	196	up
Staff resignations (quarter)	2	3	up
Share of counter staff under 6 months	44%	58%	up
Mean transaction handling time (min)	9	13	up
Customer complaints / month	60	68	up
Digital channel adoption	21%	20%	flat

WHEN IT MATTERS Horizon Bank — system variables and observed 4-month movements

Activity 5 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Map at least two reinforcing (R) and two balancing (B) loops. Which loop did the Regional Director believe he was pulling, and which loop did he actually trigger?

Q2 Month 1 showed real improvement. Explain the DELAY: why did the harmful loop take three months to overwhelm the helpful one, and what does that mean for how we evaluate quick wins?

Q3 Follow the back-office redeployment all the way round. Trace how a decision about counters ended up increasing teller resignations.

Q4 Digital adoption stayed flat at 20% throughout. Which loop is NOT running that should be, and why is a dormant balancing loop as dangerous as an active reinforcing one?

Q5 Name two leverage points. What makes a leverage point different from simply doing more of the same thing harder?

Activity 5 — Debrief

TRAINER DEBRIEF

WHAT ACTUALLY HAPPENED

He pulled B1 and unknowingly powered R1 and R2. R-loops compound, so they eventually overwhelm a one-off capacity increase.

THE DELAY IS THE TRAP

B1 acts in days; R1 and R2 act over months. That is why month 1 lied.

THE DORMANT LOOP

B2 (digital migration) was available all along and nobody activated it. Ask 'which balancing loop is NOT running?'

TOPIC 1 COMPLETE

5 Whys gave depth, Fishbone breadth, Pareto priority, Loops dynamics. You now have the full diagnostic set before touching solutions.

Lunch Break

1 hour

End of Day 1

See you at 9.30am tomorrow

RECAP

Day 1 in one line.

You cannot solve what you have not defined, and you cannot fix what you have not diagnosed.

02

TOPIC 2

Developing Corrective Plans and Evaluating Potential Solutions

Divergent ideation · SCAMPER · Impact–Ease matrix · Decision matrix · Corrective action plans

THE CENTRAL DISCIPLINE OF TOPIC 2

Generate and judge at the same time, and you will kill your best ideas.

Divergence and convergence are different cognitive modes. Run them in sequence, never in parallel.

THE SOLUTION FUNNEL

Divergence Then Convergence

1 → 2 → 3 → 4 → 5

1. Diverge
Generate widely

2. Cluster
Group by theme

3. Screen
Impact vs Ease

4. Score
Weighted matrix

5. Commit
Portfolio of four

WHY THE ORDER MATTERS

Teams that evaluate while generating stop at 6-8 conventional ideas. The uncomfortable ideas are where fixedness breaks.

Techniques for Generating Solutions

1 **Brainstorming**
All ideas welcome, judgement suspended. Quantity first, quality later.

2 **SCAMPER**
Substitute · Combine · Adapt · Modify · Put to another use · Eliminate · Reverse.

3 **Generic Parts Technique**
Strip a problem to abstract functions — solutions appear in unexpected places.

4 **Analogy / cross-industry**
Borrow a working pattern from a completely different sector.

5 **Reverse brainstorming**
Ask how to make it WORSE, then invert every answer.

6 **Nominal group technique**
Silent individual generation first, then group discussion — beats groupthink.

7 **Six Thinking Hats**
Examine the problem from facts, feelings, risks, benefits, creativity and process.

8 **Mind mapping**
Radiate ideas from the central problem to surface relationships.

BREAKING COGNITIVE FIXEDNESS

SCAMPER — Seven Moves on a Fixed Assumption

1. **Substitute**
   What can be swapped out? Different person, material, place or rule.
2. **Combine**
   What can be merged? Two steps, two roles, two services.
3. **Adapt**
   What works elsewhere that we can borrow?
4. **Modify**
   What if we magnify, shrink, or change the shape of it?
5. **Put to another use**
   What else could this resource or step do?
6. **Eliminate**
   What if we simply removed it? Often the biggest unlock.
7. **Reverse**
   What if we ran it backwards, or swapped who does what?

SCAMPER IN ACTION

The 07:00 window is fixed only because draws happen at the centre.

Reverse and Eliminate dissolve the constraint instead of working inside it. Most 'fixed' constraints are habits.

THE DISTINCTION THAT UNLOCKS OPTIONS

Regulation vs Habit — Telling Hard Constraints Apart

	Hard constraint	Habit
Meridian example	MOH phlebotomy competency standard	07:00-09:30 draws at the centre
Source	Law, regulation, contract, physics	Precedent, convenience, 'we've always'
Test	Who enforces it, and what is the penalty?	Who would actually object, and why?
If you break it	Legal or clinical consequence	Someone has to change a routine
What to do	Design within it	Challenge it explicitly

WHEN IT MATTERS Teams routinely treat habits as regulations. Ask 'who enforces this?' every single time.

TOPIC 2 · PROMPT 6

The Divergent Ideation Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as an innovation consultant running a divergent ideation session.
Do NOT evaluate or filter yet.

Root cause to solve:
<<State your diagnosed root cause>>

Constraints:
<<Budget, regulatory, contractual>>

Generate 20 distinct solutions grouped under:
- People & Retention      - Training & Onboarding
- Process & Scheduling    - Technology & Automation
- Customer/Client Experience

Rules:
- Include at least 4 ideas normally dismissed as unrealistic
- Include at least 2 borrowed from a DIFFERENT industry (say which)
- One line each, phrased as an action
- Do NOT rank, cost or evaluate

Then apply SCAMPER to this assumption:
<<State the assumption everyone treats as fixed>>
Give one idea per SCAMPER letter.
```

WATCH FOR 'Do not evaluate yet' is the most important line. Without it the AI self-censors to safe ideas.

Activity 6 — Divergent Ideation with GenAI (Breadth Before Judgement)

ACTIVITY 6

A5 · K4 — Generate a wide solution set using appropriate problem-solving techniques.

Meridian Health — solving the retention root cause

Generate 20+ solutions to the diagnosed root cause, then break a fixed assumption with SCAMPER.

TEAM

Teams of 3-5

TIME

45 minutes

TOOL

Pinboard · https://alfredang.github.io/pinboard/

The Situation

CASE STUDY

Meridian Health — solving the retention root cause

Your 5 Whys in Activity 2 established the root cause at Meridian Health: a workforce retention and onboarding failure that leaves 40% of the 07:00 roster staffed by untrained relief phlebotomists at 14 minutes per draw instead of 6.

The Clinical Director has accepted the analysis and killed the S$1.2m seventh-centre proposal. She now wants options — and she has been explicit: "Don't bring me three sensible ideas. Bring me twenty, including the ones that sound ridiculous, and then tell me which four are real."

Budget guidance: up to S$250,000 over 12 months. Constraint: MOH clinical protocols on phlebotomy competency cannot be relaxed under any circumstances.

ACTIVITY 6 · CONTEXT

The Constraints

Constraint	Detail
Budget	Up to S$250,000 over 12 months
Clinical	MOH phlebotomy competency standards are non-negotiable
Contract	07:00-09:30 fasting draw window is contractual with corporate clients
Courier	Lab courier at 09:45 is fixed by the external lab partner
Workforce	Permanent phlebotomist turnover 44%/yr; relief pool is agency-supplied
Training	Current onboarding is 2 days shadowing, no structured sign-off
Market	Phlebotomist salaries are within 5% of the market median
Shift	Early shift (06:30 start) attracts no differential pay

WHEN IT MATTERS Meridian Health — solution constraints and context

ACTIVITY 6 · RUN THIS

Activity 6 — The Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as an innovation consultant running a divergent ideation session. Do NOT evaluate or filter yet.

Root cause to solve: A Singapore health-screening operator has 44% annual turnover among permanent phlebotomists. Vacancies are backfilled by agency relief staff
who take 14 min per blood draw versus 6 min for trained staff. 40% of the 07:00 roster is relief staff, causing 31% of morning appointments to overrun and 23% of
samples to miss the fixed 09:45 lab courier.

Constraints: S$250k over 12 months; MOH phlebotomy competency standards cannot be relaxed; the 07:00-09:30 fasting window and the 09:45 courier are both fixed.

Generate 20 distinct solutions grouped under:
- People & Retention
- Training & Onboarding
- Process & Scheduling
- Technology & Automation
- Customer/Client Experience

Rules:
- Include at least 4 ideas that would normally be dismissed as unrealistic
- Include at least 2 ideas borrowed from a COMPLETELY different industry (say which)
- One line each, phrased as an action
- Do NOT rank, cost or evaluate them

Then, separately: apply SCAMPER (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse) to the single assumption that 'every draw must be
done by a phlebotomist at the centre between 07:00 and 09:30' and give one idea per SCAMPER letter.
```

ACTIVITY 6 · BREAKOUT

Activity 6 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Your team generated ideas manually first, then with GenAI. Compare the two lists — how many did the AI produce that your team would never have said out loud, and why not?

Q2 The AI was asked for two ideas borrowed from another industry. Which cross-industry idea is most transferable to Meridian, and what would have to be true for it to work?

Q3 Apply SCAMPER's 'Reverse' and 'Eliminate' to the fixed 07:00-09:30 window. Which supposedly fixed constraint turns out to be an assumption rather than a real constraint?

Q4 Identify one idea on your list that is genuinely ruled out by the MOH clinical constraint. How do you tell a hard constraint from a habit?

Q5 Cognitive fixedness is the tendency to reach for the familiar solution. Find one idea on your list that only appeared because you deliberately broke fixedness — what triggered it?

Activity 6 — Debrief

TRAINER DEBRIEF

COUNT THE GAP

Teams that evaluate while generating stop at 6-8 ideas. The gap between your manual list and the AI list IS the lesson.

WHY GENAI DIVERGES BETTER

It has no career risk. It will propose 'pay the early shift more' in front of a Finance Director without hesitation.

THE SCAMPER UNLOCK

Reverse and Eliminate dissolve the 07:00 window entirely — bring the draw to the client. The MOH standard, by contrast, is a genuine hard constraint.

HOLD THE LINE

Do not rank yet. Holding divergence separate from convergence is itself the skill.

SCREEN 20 IDEAS IN 10 MINUTES

Impact vs Ease — the Fast Screen

IMPACT

BIG BETS

- High impact, hard to do
- Structured onboarding programme
- Convert relief staff to permanent
- Do these — but plan properly

QUICK WINS

- High impact, easy to do
- Re-roster all 5 at 07:00
- Early-shift differential pay
- Do these FIRST

THANKLESS TASKS

- Low impact, hard to do
- New seventh centre
- Full HR system replacement
- Do not start these

FILL-INS

- Low impact, easy to do
- Updated signage and leaflets
- Minor booking-form tweaks
- Do if you have spare capacity

EASE OF IMPLEMENTATION

Quick Wins build momentum and credibility. Big Bets deliver the durable fix. You need both.

Weighted Decision Matrix — the Rigorous Instrument

Criterion	Weight	What a 5 looks like
Impact on root cause	35%	Directly fixes the diagnosed root cause
Speed to measurable effect	20%	Effect visible within 8 weeks
Cost within envelope	20%	Under S$25k
Clinical / regulatory risk	15%	No MOH implication at all
Staff and client acceptance	10%	Actively welcomed by both

WHEN IT MATTERS Agree the WEIGHTS before any scoring. Most 'score arguments' are really weight disagreements.

SENSITIVITY TESTING

If doubling one weight flips your winner, your recommendation is fragile.

Run the sensitivity test before you present, not after you are challenged. Boards trust robust answers.

USE THEM IN SEQUENCE

Two Tools, Two Jobs

	Impact-Ease matrix	Weighted decision matrix
Speed	Minutes	30-45 minutes
Use on	All 20 ideas	The surviving 6-8
Output	Four quadrants	A ranked, scored list
Defensibility	Directional	Auditable — you can show a board
Risk if used alone	Cannot justify the call	Wastes the session on weak ideas

WHEN IT MATTERS Screen with the light tool; decide with the heavy one.

TOPIC 2 · PROMPT 7

The Convergence Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a decision analyst. I will give you a list of candidate solutions. Do NOT add new ideas.

STEP 1 — Classify every solution on an Impact vs Ease matrix as exactly one of:
- Quick Win (high impact, easy)
- Big Bet (high impact, hard)
- Fill-In (low impact, easy)
- Thankless Task (low impact, hard)
Give a one-line reason for each placement.

STEP 2 — Take only the Quick Wins and Big Bets and score them in a weighted decision matrix:
- Impact on root cause (weight 35%)
- Speed to measurable effect (20%)
- Cost within a S$250k envelope (20%)
- Clinical/regulatory risk (15%)
- Staff and client acceptance (10%)
Score each 1-5, show the weighted total, and rank them.

STEP 3 — Recommend the best FOUR as a portfolio, and explicitly state:
- why this combination is stronger than the top four individual scores
- which two solutions overlap or make each other redundant
- what the portfolio still leaves unaddressed

Solutions: [paste your team's list here]
```

WATCH FOR The AI invents cost figures with total confidence. Its ranking is only as good as your criteria.

Activity 7 — Converging: Impact-Ease Matrix and Weighted Decision Matrix

ACTIVITY 7

A5 · K4, K6 — Shortlist and evaluate the most viable ideas using decision-making techniques.

Meridian Health — choosing four from twenty

Screen twenty ideas down to four fundable solutions — and defend the ranking with evidence.

TEAM

Teams of 3-5

TIME

50 minutes

TOOL

Pinboard · https://alfredang.github.io/pinboard/

The Situation

CASE STUDY **Meridian Health — choosing four from twenty**

You now hold twenty-plus candidate solutions from Activity 6. The Clinical Director will fund approximately four, inside S$250,000 over 12 months.

Two members of your team are already advocating loudly for their favourite ideas and the discussion is circling. This is exactly the moment where teams either make an evidence-based decision or default to whoever is most senior or most persistent.

Your job is to convert the board into a defensible shortlist using two complementary tools: a fast Impact-Ease screen, then a weighted decision matrix on the survivors.

ACTIVITY 7 · CRITERIA

The Decision Criteria

Criterion	Weight	Scoring guidance (1-5)
Impact on root cause (retention/training)	35%	5 = directly fixes the root cause
Speed to measurable effect	20%	5 = effect visible within 8 weeks
Cost within S$250k envelope	20%	5 = under S$25k
Clinical/regulatory risk	15%	5 = no MOH implication at all
Staff and client acceptance	10%	5 = actively welcomed by both

WHEN IT MATTERS Meridian Health — agreed decision criteria and weightings

Activity 7 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Place all your ideas on the Impact-Ease matrix. Which quadrant is most crowded, and what does a crowded Fill-In quadrant tell you about how your team was thinking?

Q2 Run the weighted matrix. Did the top-scoring solution match your team's gut favourite from Activity 6? If not, what did the weighting expose?

Q3 Change the weight on 'Speed to measurable effect' from 20% to 40% and re-rank. Does the winner change? What does that sensitivity tell you about how robust your recommendation is?

Q4 The AI recommended a portfolio of four. Explain why the best four INDIVIDUAL scores are not automatically the best four TOGETHER.

Q5 Consensus fatigue is when a team accepts a weak option just to end the discussion. Where in this activity was your team most at risk of it, and what stopped you?

Activity 7 — Debrief

TRAINER DEBRIEF

THE WEIGHTS ARE THE DECISION

Surfacing weights BEFORE scoring converts a political argument into an explicit, recorded choice.

PORTFOLIO, NOT LEADERBOARD

The top four scores may all attack the same cause, or be redundant. Balance quick stabilisation against durable repair.

SENSITIVITY

If the winner holds under several weightings, you have a robust recommendation. If it flips, say so.

THE AI'S LIMIT

It is a fast, tireless scorer with no politics — and no accountability. The Clinical Director will ask YOU why, not the model.

The Corrective Action Plan — Eight Components

1 Root cause addressed
Ties the action to the diagnosis. Without it, solutions drift to symptoms.

2 Corrective action
The specific thing being done — one sentence, unambiguous.

3 Owner (named role)
Single point of accountability. A department cannot be phoned.

4 Timeline / milestone
Start, milestone, completion — in weeks, not quarters.

5 Resources required
Budget, people, tools — committed, not hoped for.

6 Success measure
Metric + baseline + target. This is what makes evaluation possible.

7 Risk and mitigation
What breaks it, and the action (with an owner) that prevents that.

8 Review checkpoint
A date AND the decision to be made: continue, extend or stop.

THE OWNER IS THE SPINE

'Operations' cannot be phoned, held to a date, or asked why it slipped.

One named owner per action. No co-owners. This single discipline prevents the most common cause of project death.

Measures That Work vs Measures That Don't

	Unmeasurable	Measurable with existing data
Morale	'Improved staff morale'	Voluntary resignations per quarter
Capability	'Better trained staff'	Mean relief-staff draw time: 14 → under 8 min
Quality	'Fewer mistakes'	First-attempt success rate: 82% → 92%
Coverage	'Properly staffed mornings'	07:00 roster fill rate by permanent staff

WHEN IT MATTERS If the organisation cannot measure it today, either change the measure or add a task to start collecting it.

TOPIC 2 · PROMPT 8

The Corrective Action Plan Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
You are an execution-focused planning assistant. Convert each approved solution into a corrective action plan row. Be concrete and brief — no theory.

For EACH solution, produce exactly these fields:
- Root cause addressed (link back to the diagnosis)
- Corrective action (one specific sentence)
- Owner (a job title, not a department)
- Timeline (start, key milestone, completion)
- Resources (budget in S$, people, tools)
- Success measure (metric + baseline + target)
- Key risk + mitigation
- Review checkpoint (date and the decision to be made at it)

Then flag, in a separate section:
1. Any two actions that depend on each other and must be sequenced
2. Any owner who appears on more than two actions (overload risk)
3. Any success measure that cannot actually be measured with data the organisation already has

Approved solutions: [paste your four here]
Root cause: 44% phlebotomist turnover leaving 40% of the 07:00 roster untrained, causing 14-min draws vs 6-min, 31% appointment overruns and 23% of samples missing the 09:45 courier.
Budget: S$250,000 over 12 months.
```

WATCH FOR Asking the AI to flag dependencies and overloaded owners catches what a close-in team misses.

Activity 8 — Building the Corrective Action Plan

ACTIVITY 8

A4 · K9 — Develop corrective action plans for shortfalls identified.

Meridian Health — from chosen solutions to an auditable plan

Convert four approved solutions into an auditable plan another team could execute without you.

TEAM

Teams of 3-5

TIME

45 minutes

TOOL

Pinboard · https://alfredang.github.io/pinboard/

The Situation

CASE STUDY **Meridian Health — from chosen solutions to an auditable plan**

The Clinical Director has approved your four solutions. She now asks the question that separates a good analysis from a delivered result:

"Who is doing what, by when, with what money, and how will I know it worked?"

She also adds a warning from experience: "The last improvement project we ran had a beautiful slide deck and no owner. Six months later nothing had changed and everyone assumed someone else was doing it."

Your team must convert four approved solutions into a corrective action plan that would survive an internal audit — and that a colleague could execute without you in the room.

ACTIVITY 8 · TEMPLATE

The Components

Component	Why it exists	Failure if missing
Root cause addressed	Ties the action to the diagnosis	Solution drifts to a symptom
Corrective action	The specific thing being done	Vague intent, no execution
Owner (named person)	Single point of accountability	Everyone assumes someone else
Timeline / milestone	Makes progress checkable	Slips indefinitely, unnoticed
Resources required	Budget, people, tools committed	Stalls at first resource conflict
Success measure + baseline	Defines what 'worked' means	Cannot evaluate; endless debate
Risk and mitigation	Anticipates what breaks it	Surprised by predictable failure
Review checkpoint	Forces a decision to continue/stop	Runs on past the point of failure

WHEN IT MATTERS Corrective Action Plan — mandatory components (K9)

Activity 8 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Complete all eight components for each of your four actions. Which component did your team find hardest to fill honestly, and what does that reveal?

Q2 Every owner must be a named role, not a department. Why does 'Operations' fail as an owner where 'Centre Operations Manager' succeeds?

Q3 Check your success measures against the baseline in your original problem statement. Can each one actually be measured with data Meridian already collects? Which cannot?

Q4 The AI flagged dependencies between actions. Which two of your actions must be sequenced, and what breaks if you run them in parallel?

Q5 Identify the biggest risk across your whole plan. Is your mitigation a real action with an owner, or a hope?

Activity 8 — Debrief

TRAINER DEBRIEF

THE BASELINE PAYS OFF

Teams that skipped the baseline in Activity 1 literally cannot write the success measure now. The course has closed its own loop.

SEQUENCING

Converting relief staff to permanent DEPENDS on the onboarding programme existing — otherwise you lock in the defect.

OWNER OVERLOAD

If one role owns three of four actions, the plan is a single point of failure however good each row looks.

CHECKPOINTS CARRY DECISIONS

'Review progress' is theatre. 'Decide to roll out, extend or stop' is governance.

Lunch Break

1 hour

03

TOPIC 3

Selecting, Implementing and Measuring Solution Effectiveness

Implementation planning · Stakeholders & resistance · Change management · KPIs · PDCA

K8 · WHAT ACTUALLY DETERMINES SUCCESS

The CRM worked perfectly. That was not the problem.

Solutions fail on adoption far more often than on design. Implementation is mostly about people.

THE IMPLEMENTATION ARC

From Decision to Proven Result

1. **Sequence** — What before what
2. **Assign** — Named owners
3. **Engage** — Stakeholders & resistance
4. **Execute** — With checkpoints
5. **Evaluate** — Against the baseline

THE CLOSING LOOP

Evaluation compares against the baseline captured in your problem statement. Topic 3 only works because Topic 1 was done properly.

Factors Affecting Implementation Effectiveness

1. **Sponsorship**
   A senior owner who can re-prioritise competing work — not just endorse it.

2. **Sequencing**
   Dependencies respected. A recovery campaign before a speed fix burns the customer twice.

3. **Capacity**
   Teams already at capacity will not absorb new work by goodwill.

4. **Stakeholder buy-in**
   Resistance is a design input, not a surprise to be managed later.

5. **Communication**
   Regular, specific, and continuing past launch — not a single kick-off email.

6. **Measurement**
   Defined at day 0, with the window and the control agreed before you start.

SECURING ADOPTION

Change Management Actions That Work

1

Leadership alignment

Communicate the business goal and tie measurable KPIs to outcomes.

2

Quick wins first

Deliver an early visible improvement to build momentum and credibility.

3

Involve early

Bring affected teams into solution design before rollout, not after.

4

Co-design with resisters

An imposed standard is complied with on paper. A co-designed one sticks.

5

Incentives & accountability

Link the change to performance metrics people are actually measured on.

6

Train and support

SOPs, quick guides and a real helpdesk — adoption dies without support.

RESISTANCE IS INFORMATION

Reading Stakeholder Resistance

Stakeholder type	What they actually object to	The move that works
At-capacity team	Workload, not the idea	Real relief — re-prioritise or fund capacity
Face at risk	Being shown to have been wrong	Reframe; give them ownership of a win
External party	Cost and scrutiny	Co-design, then tie to commercial consequence
Job-security fear	A rational threat	Commit only to what you can guarantee
The indifferent	Nothing — and that is the danger	Give them a stake before you need them

WHEN IT MATTERS The messenger matters as much as the message. Assign one to every conversation.

THE QUIET RISK

Indifference kills more rollouts than opposition.

Nobody escalates an absence of enthusiasm — so it is never fixed until the launch fails.

TOPIC 3 · PROMPT 9

The Implementation & Change Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a change management and implementation planning advisor.

Approved solutions:
<<List your solutions>>

PART A — Implementation plan. For each solution give ONLY:
- Action (what will be done)
- Method/tool (how)
- Owner and timeline (who, by when)
- Sequencing note (what must happen before it)

PART B — Stakeholder strategy. For EACH stakeholder give:
- Their specific objection in THEIR words, not yours
- What they need to hear or receive to move
- One concrete action to secure cooperation
- Who should deliver that message, and why

PART C — Name the ONE stakeholder most likely to sink this
rollout and explain what makes them dangerous. Then give a
90-day communication plan: what, to whom, how often, by whom.

Be blunt about political realities. Do not assume goodwill.
```

WATCH FOR The AI has no stake in your office politics — it will name the face-saving problem you would not write down.

Activity 9 — Implementation Planning and Stakeholder Resistance

ACTIVITY 9

A6 · K3, K8 — Draw up implementation plans and address factors affecting effectiveness.

ShopFront SG — the rollout that has to survive contact with people

Build the rollout plan AND the stakeholder strategy that makes it survive contact with people.

TEAM

Teams of 3-5

TIME

50 minutes

TOOL

Pinboard · https://alfredang.github.io/pinboard/

The Situation

CASE STUDY

ShopFront SG — the rollout that has to survive contact with people

Return to ShopFront SG from Activity 1. The diagnosis is complete and the solution set is approved: a one-page guest checkout, a CDN and caching layer to bring app load under 2 seconds, a supplier QC checklist to cut the return rate, and cart-recovery notifications.

The Chief Operating Officer has seen improvement projects die before. She says:

"The plan is fine. Now tell me who is going to fight it, why, and what you are going to do about it. Last year we bought a S$400,000 CRM that nobody used. It worked perfectly. That was not the problem."

Your team must build the implementation plan AND the stakeholder strategy that makes it stick. Assume nothing about goodwill.

The Stakeholders

Stakeholder	Interest	Likely position
IT Development team	Already at capacity on a POS migration	Resist — sees added workload
Head of Marketing	Sponsored the S$180k discount campaign	Resist — solution implies her spend was wrong
Store Operations	In-store sales flat, feels unaffected	Indifferent — 'not my problem'
Suppliers (12 vendors)	Face new QC standards and possible delisting	Resist — cost and scrutiny
Customer Service	Handling 410 tickets/week, overloaded	Support — expects relief
Finance	Approved budget, wants ROI evidence	Conditional — needs measurement
COO (sponsor)	Accountable for the turnaround	Champion
Frontline retail staff	Fear digital shift reduces store headcount	Anxious — job security

WHEN IT MATTERS ShopFront SG — stakeholder map and known positions

Activity 9 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Build the implementation plan for all four solutions. Which two MUST be sequenced rather than run in parallel, and what breaks if you ignore that?

Q2 The Head of Marketing resists because the diagnosis implies her S$180k campaign was misdirected. This is a face problem, not a logic problem. How do you secure her cooperation without requiring her to admit error?

Q3 Frontline retail staff fear the digital push costs them jobs. Is that fear irrational? What would you actually commit to — and what happens to your credibility if you promise something you cannot guarantee?

Q4 Twelve suppliers face new QC standards. What is the difference between imposing the checklist and getting them to adopt it, and which one survives after month three?

Q5 Identify the single stakeholder most likely to sink this rollout. Note: it may not be the loudest one. Justify your choice.

Activity 9 — Debrief

TRAINER DEBRIEF

SEQUENCING IS NOT OPTIONAL

Cart recovery must follow the speed fix. Driving traffic back to a 4.8-second app burns the customer twice.

THE FACE PROBLEM

Marketing's objection is reputational, not logical. Reframe rather than relitigate — and let the COO deliver it, peer to peer.

CO-DESIGN BEATS IMPOSITION

An imposed supplier checklist is complied with on paper for two months. Watch what happens to it in Activity 10.

RESISTANCE IS DATA

IT's objection revealed a capacity conflict the plan had not costed. That improves the plan.

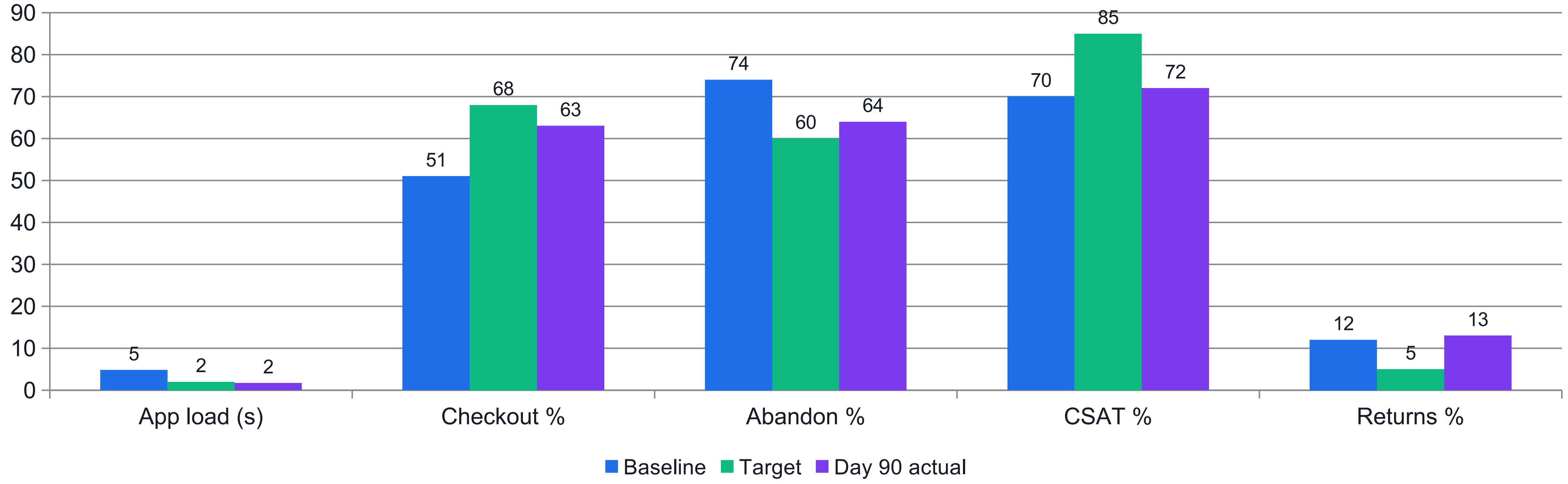
Leading vs Lagging Indicators

	Leading	Lagging
Responds in	Days to weeks	Months to quarters
ShopFront example	App load, checkout completion, adoption	CSAT, repeat purchase, brand trust
Tells you	Whether the change is taking hold	Whether the outcome actually improved
Risk of using alone	Early signal, may not persist	Too late to steer
Planning implication	Steer with these	Set realistic horizons for these

WHEN IT MATTERS Expecting CSAT to move in 90 days was a planning error, not an execution failure.

DID IT WORK?

ShopFront SG — 90 Days Against Baseline



WHAT THE DATA SHOWS

App load beat target. Checkout and abandonment improved but fell short. CSAT barely moved — it is a lagging indicator. Returns got WORSE, which needs explaining before anyone concludes the solution failed.

Evaluating Honestly — the Four Questions

1

Did the metric move?

Against the BASELINE and the TARGET — both, with numbers.

2

Can you attribute it?

Or is it confounded by season, market or another change? Say which.

3

Solution or implementation?

A solution not adopted has not been tested. Those need opposite responses.

4

What would change your mind?

Naming this is what makes your recommendation credible to a CFO.

DECOMPOSE BEFORE YOU CONCLUDE

The return rate got worse — and the solution did not fail.

Only 4 of 12 suppliers adopted the checklist, while higher volume brought in first-time buyers who return more. That is a mix shift, not a quality collapse.

THE DISTINCTION THAT DECIDES THE NEXT S$200K

Solution Failure vs Implementation Failure

	Solution failure	Implementation failure
What happened	It was done, and it did not work	It was never really done
ShopFront example	Cart recovery at 3.1% vs 8% target	Supplier QC — 4 of 12 adopted
Evidence to check	Open rates, click rates, benchmarks	Adoption and compliance rates
Right response	Change or drop the solution	Fix the rollout and change management
Traceable back to	Weak diagnosis or wrong target	Stakeholder resistance you predicted

WHEN IT MATTERS Abandoning a solution that was never adopted throws away a fix that might have worked.

TOPIC 3 · PROMPT 10

The Evaluation Prompt

PROMPT · Paste into ChatGPT · Copilot · Gemini · Claude

```
Act as a performance evaluation analyst. Assess whether this improvement programme worked. Be rigorous and do not flatter the results.

Results after 90 days (metric: baseline -> target -> actual):
App load 4.8s -> <2.0s -> 1.7s
Checkout completion 51% -> 68% -> 63%
Cart abandonment 74% -> 60% -> 64%
Online sales S$1.9m -> S$2.8m -> S$2.5m
CSAT 70% -> 85% -> 72%
Complaints/week 410 -> 150 -> 295
Product return rate 12% -> 5% -> 13% (WORSE)
Repeat purchase 29% -> 40% -> 31%
Supplier QC checklist adopted by 4 of 12 suppliers
Cart recovery conversion 3.1% against an 8% target

Context: the 90-day window included the Great Singapore Sale.

Produce:
1. A verdict per metric: Met / Partially met / Not met / Deteriorated
2. Which solutions can be credited with which movements — and where you CANNOT establish attribution, say so explicitly and explain why
3. Why the product return rate got WORSE despite the QC initiative — give the most likely explanation and state what evidence would confirm it
4. Separate LEADING from LAGGING indicators here, and explain why CSAT and repeat purchase barely moved even though operational metrics improved
5. How the Great Singapore Sale confounds this evaluation, and what analysis would isolate the programme effect from the seasonal effect
6. A clear recommendation to the CFO on releasing the next S$200k tranche: continue, adjust, or stop — with reasoning
7. The three things you would measure differently next time
```

WATCH FOR Without an explicit instruction to flag what cannot be determined, the AI will quietly overclaim attribution.

Activity 10 — Measuring Effectiveness: Did It Actually Work?

ACTIVITY 10

A7 · K7 — Evaluate the effectiveness of implemented solutions and implementation plans.

ShopFront SG — 90 days later, and the numbers are ambiguous

Judge an ambiguous 90-day result honestly and make the funding call to the CFO.

TEAM

Teams of 3-5

TIME

45 minutes

TOOL

Pareto Chart Tool · https://alfredang.github.io/paretochart/

The Situation

CASE STUDY

ShopFront SG — 90 days later, and the numbers are ambiguous

Ninety days have passed. The COO has the results and she is not sure what to make of them. Some numbers moved well. Some did not move at all. One moved in the wrong direction. And the quarter included the Great Singapore Sale, which everyone is now using to explain whichever result suits their argument.

The Head of Marketing says the improvement is obviously seasonal. The IT lead says the improvement is obviously the CDN. Finance wants to know whether to release the next S$200,000 tranche.

Your team must evaluate honestly: what worked, what did not, what you cannot yet tell, and what you recommend next. "It's working" is not an acceptable answer.

ACTIVITY 10 · DATA

The Results

Metric	Baseline	Target	Day 90 actual	Verdict
App page load	4.8 s	under 2.0 s	1.7 s	Target met
Checkout completion	51%	68%	63%	Improved, short
Cart abandonment	74%	60%	64%	Improved, short
Online sales / month	S$1.9m	S$2.8m	S$2.5m	Improved, short
CSAT	70%	85%	72%	Barely moved
Complaint tickets / week	410	150	295	Improved, short
Product return rate	12%	5%	13%	WORSE
Repeat purchase (90-day)	29%	40%	31%	Barely moved
Guest checkout adoption	n/a	n/a	44% of orders	New data
Supplier QC checklist adoption	0	12 of 12	4 of 12	Behind
Cart recovery emails sent	n/a	n/a	18,400	Live
Cart recovery conversion	n/a	8%	3.1%	Underperforming

WHEN IT MATTERS ShopFront SG — 90-day results against baseline

Activity 10 — Discussion Questions

DISCUSSION QUESTIONS

Q1 Give a verdict on each metric. Overall — did this programme succeed? Defend a single-sentence answer to the COO.

Q2 The product return rate got WORSE (12% to 13%) despite the QC initiative. Give the most likely explanation, and say what evidence would confirm or refute it.

Q3 App load smashed its target but CSAT barely moved. Explain this using leading versus lagging indicators. How long would you expect CSAT to lag?

Q4 The Great Singapore Sale sits inside the measurement window. How would you isolate the programme effect from the seasonal effect? What would you have done differently at day 0 to make this answerable?

Q5 Cart recovery converts at 3.1% against an 8% target. Is the solution wrong, or the implementation, or the target? How would you tell the difference — and what does each answer imply?

Q6 Make the call to the CFO: release the next S$200,000, adjust, or stop. Justify it with the evidence, and name what would change your mind.

Activity 10 — Debrief

TRAINER DEBRIEF

ATTRIBUTION IS THE SKILL

App load is cleanly attributable. Online sales are confounded by the GSS and cannot be claimed either way. Say so — that honesty is what makes the rest credible.

THE RETURNS PUZZLE

Mix shift, not quality collapse. Segment by supplier and by customer type to confirm before concluding anything.

THE CONNECTION THAT MATTERS

Supplier QC missed because suppliers resisted — exactly as predicted in Activity 9. Your stakeholder analysis foretold your missed metric.

THE VERDICT

Continue with adjustment: rebase the cart-recovery target, fix supplier adoption as a change-management problem, and give CSAT a longer horizon.

Sustaining the Gain

1 **PDCA / DMAIC-Control**
Plan-Do-Check-Act. Without a control phase, improvements decay back.

2 **Standardise**
Write the new way into the SOP, the induction and the checklist.

3 **Monitor**
Keep the leading indicator on a dashboard someone actually looks at.

4 **Assign ownership**
A named role owns the sustained metric after the project team disbands.

5 **Review cadence**
A standing checkpoint with a decision, not a status update.

6 **Capture the lesson**
'Look back and learn' — the L in IDEAL, and the step most often skipped.

WRAP UP

Summary, Q&A and Assessment

TWO DAYS IN ONE DIAGRAM

The Complete Method You Now Have

1. **Frame** — Six-element statement
2. **Diagnose** — 5 Whys · Fishbone ·…
3. **Generate** — Diverge widely, SCAMPER
4. **Decide** — Impact-Ease then weighted…
5. **Deliver** — CAP · stakeholders ·…

THE ONE THING TO REMEMBER

GenAI expands what you consider and accelerates how you get there. The evidence, the judgement and the accountability remain yours.

KEEP USING THESE AT WORK

Your Problem-Solving Toolkit

1 **5 Whys**
https://alfredang.github.io/5whys/

2 **Fishbone**
https://alfredang.github.io/fishbone/

3 **Pareto Chart**
https://alfredang.github.io/paretochart/

4 **System Loop**
https://alfredang.github.io/systemloop/

5 **CollabNote**
https://alfredang.github.io/collabnote/

6 **Pinboard**
https://alfredang.github.io/pinboard/

THE PROMPT LIBRARY IN YOUR LEARNER GUIDE

Ten Prompts You Can Reuse Tomorrow

1

1 · Problem statement
Refine a vague complaint into six labelled elements.

2

2 · 5 Whys facilitator
One why at a time, demanding evidence at each level.

3

3 · Fishbone populator
Six bones, each cause marked [EVIDENCED] or [HYPOTHESIS].

4

4 · Pareto analyst
Cumulative percentages and the vital-few cut-off.

5

5 · Systems mapper
R and B loops, the delay, and the leverage points.

6

6 · Divergent ideation
20 ideas plus SCAMPER on a fixed assumption.

7

7 · Decision analyst
Impact-Ease screen then a weighted matrix.

8

8 · CAP builder
Eight components, with dependency and overload flags.

9

9 · Change advisor
Objections in their words, plus the right messenger.

10

10 · Evaluator
Verdicts, attribution, and what cannot be determined.

BEFORE WE ASSESS

Summary & Q&A

What is the one problem you will go back and re-frame on Monday?

BEFORE YOU LEAVE

Certificate & TRAQOM Survey (Mandatory)

1

Complete the survey

The TRAQOM course quality survey is mandatory for WSQ-funded courses.

2

Where

https://lms-tms.tertiaryinfotech.com

3

Certificate

Your Certificate of Achievement is released after the survey and a Competent result.

4

SSG record

Your Statement of Achievement appears in your SkillsFuture / MySkillsFuture record.

READ BEFORE THE ASSESSMENT

Briefing for Assessment

1. **Clear your table**
   Phones and materials under the table or on the floor.

2. **No photos or recording**
   Assessment scripts must not be photographed or copied.

3. **No discussion**
   Individual work only once the paper is handed out.

4. **Black or blue pen**
   No liquid paper or correction tape on hard-copy scripts.

5. **Time is called**
   Scripts are collected when time is up.

6. **Open book means**
   Slides, Learner Guide and approved materials — no internet, no AI tools.

Final Assessment

1

Written Assessment (WA)

Short-answer questions covering problem identification, root cause analysis, solution selection and evaluation.

2

Case Study (CS)

An integrated workplace scenario — you diagnose, decide and justify, exactly as in the activities.

3

Open book

Slides, Learner Guide and approved materials only. No internet search, no AI tools during assessment.

4

Competent / Not Yet Competent

Both instruments must be assessed Competent. A re-assessment path is available.

FINAL ASSESSMENT

Assessment Flow

1. **TRAQOM Survey** — Complete on the LMS
2. **Digital Attendance** — Assessment QR scan
3. **Written Assessment** — SAQ · 1 hour
4. **Case Study** — Scenario · 1 hour
5. **Submit on LMS** — Upload your answers
6. **Summary Record** — Sign the ASR

RE-ASSESSMENT

A learner assessed Not Yet Competent is counselled on the gap and offered a re-assessment. An appeal may be lodged with the training administrator within 7 working days.

MANDATORY ADMIN

Digital Attendance & TRAQOM Survey

1

AM · PM · Assessment

Digital attendance is taken three times a day. It is mandatory for every WSQ-funded course.

2

Scan the SSG QR code

The trainer displays the QR code from the SSG portal. Scan it with your phone camera and submit.

3

75% minimum attendance

Below 75% you are not eligible for assessment or for course fee funding.

4

TRAQOM survey

Complete the mandatory post-course survey and collect your certificate at https://lms-tms.tertiaryinfotech.com

CONTINUE YOUR LEARNING

Recommended Next Courses

1. **Project Management with Generative AI (GenAI)**
   WSQ funded · tertiarycourses.com.sg

2. **Design Thinking Course for Businesses**
   WSQ funded · tertiarycourses.com.sg

3. **Effective Project Management for Small Projects**
   WSQ funded · tertiarycourses.com.sg

4. **Project Management Professional (PMP) 35 PDU Training**
   WSQ funded · tertiarycourses.com.sg

5. **Mastering Notion for Content, Project, and Database Management**
   WSQ funded · tertiarycourses.com.sg

WE'RE HERE TO HELP

Support

1

Email

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2

Telephone

+65 6100 0613

3

Website

www.tertiarycourses.com.sg

4

After the course

Free post-course consultation on subject-matter queries by email.

Thank You!

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